

MsGA-Edit: Layout-Preserving Weather Transfer in Driving Scenes via Multi-Source Guided Attention

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Abstract—Recent text-conditioned diffusion models enable strong zero-shot image editing, yet in semantically rich and structurally complex driving scenes, stronger CFG-guided weather appearance often corrupts geometric layouts and semantic structures. We attribute this failure to cross-attention leakage, where high responses to global style tokens, such as snow, rain, and fog, interfere with layout modeling. Although prior methods mitigate style–structure coupling through cross-attention manipulation, they do not compensate for structural information loss within self-attention computation, leading to persistent failures under strong weather prompts. To address this problem, we propose Multi-source Guided Attention (MsGA), which operates in self-attention layers to reduce weather-induced layout drift. Based on MsGA, we introduce MsGA-Edit, a training-free diffusion editing framework that preserves geometric integrity while performing prompt-guided weather synthesis. First, we derive structure-aware guidance by inverting the clear-weather image and its semantic and edge maps into aligned DDIM trajectories, where masked structural priors preserve critical street layouts, including roads, vehicles, and buildings. Then, MsGA injects Spatial-Feature Structural Guidance and Self-Attention Structural Guidance into text-driven generation, with depth-aware value injection further improving structure preservation in distant regions. Compared with state-of-the-art training-free diffusion editing baselines, MsGA-Edit achieves a better overall trade-off between weather appearance transfer strength and layout preservation. Using MsGA-Edit, we synthesize RoadWeather-34K, a large-scale synthetic benchmark that spans diverse road scenes, including urban, suburban, and countryside scenarios, and provides six weather/appearance styles with six CFG-guided visual appearance grades per style. Downstream evaluations on real challenging-weather data demonstrate that detectors trained or fine-tuned with our synthetic augmentation consistently improve robustness across multiple model families, achieving gains of up to +19.5 in mAP₅₀ and +22.1 in mean matched-box IoU (MBIoU). Under the ACDC synthetic-to-real protocol, detectors trained on MsGA-Edit synthetic data achieve 77.9% of the ACDC-real upper bound in mAP₅₀.

Index Terms—Autonomous driving, zero-shot image editing, diffusion models, weather effect transfer, traffic-scene synthesis.

I. INTRODUCTION

Autonomous Driving Systems (ADS) perform strongly in clear weather, yet their perception degrades markedly under challenging weather conditions such as rain, snow, fog, and sand/dust storms [1]. These phenomena obscure content, reduce sensor reliability, and alter the appearance of critical road users and infrastructure. At the same time, collecting large-scale, safety-compliant data that spans multiple weather categories and visual severity levels is expensive and logistically challenging. As a result, public benchmarks such as KITTI [2], nuScenes [3], and Cityscapes [4] are dominated by

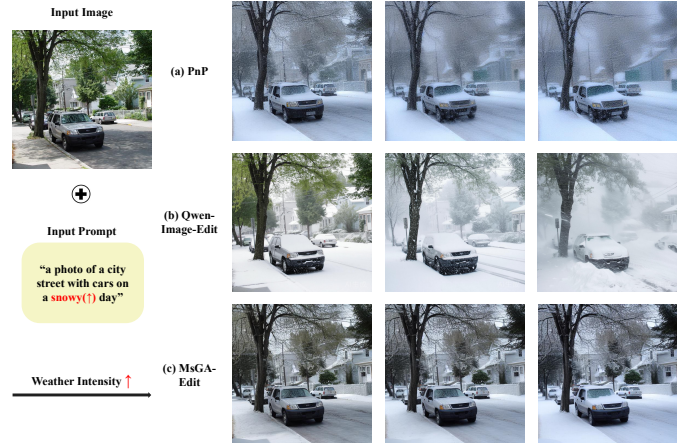


Fig. 1. **Weather-effect transfer under increasing classifier-free guidance across three editing methods.** We compare (a) Plug-and-Play Diffusion [5], (b) Qwen-Image-Edit [6], and (c) our MsGA-Edit under increasingly strong CFG-driven weather appearance from left to right. (a) and (b) suffer from *cross-attention leakage*: high weights on global style tokens leak information into layout modeling, causing structural distortion. In contrast, our approach keeps street layout and object semantics more stable as the visual weather effect increases.

clear-weather imagery, creating a distribution gap that limits robustness.

To mitigate the scarcity of challenging-weather data, a large body of work explores image synthesis and editing. Early Generative Adversarial Network (GAN) approaches [7], [8] often require paired supervision and struggle to generalize across domains. Diffusion models [9], [10] have recently emerged as a compelling alternative, enabling high-fidelity generation and zero-shot, text-guided editing. Representative editors either inject source-image features into the denoising process (e.g., PnP [5]), learn instruction-following edits from paired data (Instruct-Pix2Pix [11]), or rely on inversion to preserve content before applying edits (DiffuseIT [12], Edit-Friendly DDPM Inversion [13]). Editing-aware schedules further strengthen locality and reduce unintended changes (LEDITS++ [14]).

However, directly applying zero-shot editing to real driving scenes yields a consistent failure, as illustrated in Fig. 2 [15], [16]. As the CFG-guided weather prompt becomes stronger, token-level cross-attention coupling assigns high weights to global style tokens (snow/rain/fog), which then leak into the layout modeling stream. Recent analyses [17] have provided deeper insights into this issue, revealing that cross-attention maps in Stable Diffusion often embed object attribution and semantic information, leading to unintended entanglements where style features inadvertently influence structural ele-

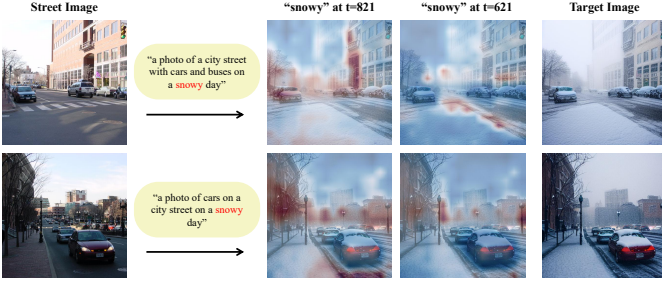


Fig. 2. **Cross-attention layer feature visualizations at timesteps 621 and 821.** Features for the prompt word “snowy” are extracted and converted to importance-weighted heatmaps as shown. The dispersed distributions vary across timesteps and interfere with street-scene structural generation. This illustrates the *cross-attention leakage* phenomenon [15], [16]: high weights on global style tokens leak style information into layout modeling.

ments. In contrast, self-attention maps primarily capture spatial associations and geometric structures, without strong semantic encoding. This disparity highlights why prior approaches, which mainly manipulate cross-attention to mitigate style-structure coupling, overlook direct intervention in the self-attention computation, resulting in persistent failures under strong weather prompts. This failure is further illustrated in Fig. 3, where geometry and semantics are corrupted through object disappearance, boundary distortions, and hallucinated structures, together with unstable edit strength (under-/over-transfer). In short, style overwhelms structure when the prompt guidance becomes stronger. To address this, our method introduces Multi-source Guided Attention (MsGA) into self-attention layers to preserve layout under style-driven weather editing by explicitly guiding the self-attention computation, drawing on the understanding that self-attention is key for preserving layout integrity [17].

Thus, we propose MsGA-Edit, a training-free, two-stage diffusion framework tailored for layout-preserving weather-effect editing in driving scenes. The first stage, named Structure-Aware Guidance Extraction, performs synchronized DDIM inversion for the original street image I^G , semantic map I^S , and Canny edge map I^E , and constructs Structure-Aware Guidance from two parts: Self-Attention Structural Guidance and Spatial-Feature Structural Guidance. The former is obtained through semantic/edge Q/K fusion followed by Self-Attention Guidance Filtering, while the latter is extracted only from the original street-image feature F^G . The second stage performs target-weather generation with DDIM denoising. During this stage, the filtered Q/K guidance and the spatial-feature guidance are injected at selected DDIM denoising steps; DVI is coupled with the Q/K-based MsGA branch to produce the depth-modulated weather value V^d whenever MsGA is active. We detail these two stages below.

Stage 1: Structure-Aware Guidance Extraction. To reduce style-induced structural drift in zero-shot weather editing, MsGA-Edit aligns three co-registered inputs—the original street image, its semantic map, and its Canny edge map—through deterministic DDIM inversion with a shared schedule. At selected self-attention layers and scheduled DDIM indices, semantic and edge Q/K carriers are cached, fused into Self-Attention Structural Guidance, and then filtered by

the structure mask M_ℓ , so only mask-related Q/K components are retained. In parallel, Spatial-Feature Structural Guidance is cached only from the original street-image feature F^G . These cached priors are not injected in Stage 1; they are used later during Stage 2 DDIM denoising.

Stage 2: MsGA with DVI and Mask-Gated Fusion. During the 50-step DDIM target denoising stage, MsGA computes structure-guided attention from the filtered Self-Attention Structural Guidance, while the Spatial-Feature Structural Guidance provides the source-preserving branch. The fixed spatial gate $W_\ell(x) = \gamma M_\ell(x)$ determines how strongly the structure-guided branch contributes inside the selected mask, while $M_\ell(x) = 0$ outside the mask makes the output fall back to the original street-image feature branch. DVI is coupled with the MsGA Q/K injection window: when the Q/K-guided MsGA branch is active, it converts the weather value V^w into V^d according to the depth sensitivity β and the DVI strength s_{DVI} . This formulation keeps the intervention in self-attention, avoids direct cross-attention manipulation, and gives each parameter a clear role.

We validate MsGA-Edit under strictly zero-shot settings on KITTI, nuScenes, and Cityscapes [2]–[4], where it achieves a better overall trade-off between weather appearance transfer and layout preservation than state-of-the-art training-free diffusion editing baselines. Leveraging CFG-guided appearance adjustment, we further synthesize *RoadWeather-34K*, a large-scale synthetic benchmark spanning diverse road scenes (e.g., urban, suburban, and countryside) with six weather/appearance styles and six CFG-guided visual appearance grades per style. Finally, we demonstrate practical downstream benefits on real challenging-weather perception. Starting from RoadWeather-34K, we curate a high-quality strong-weather training subset (rain/snow/fog/sandstorm) by filtering out clear and weak-effect samples and removing failed generations. We fine-tune three modern detectors (YOLOv8 [18]/YOLOv11 [19]/RF-DETR [20]) using only this synthetic subset, and evaluate them zero-shot on the real DAWN benchmark [21] without using any DAWN labels. Across model families and scales, we observe consistent robustness gains: mAP₅₀ improves by up to +19.5 points and mean matched-box IoU (MBIoU) by up to +22.1 points, accompanied by a substantial reduction in Miss (up to 26 points). In the ACDC synthetic-to-real protocol, MsGA-Edit also improves over GAN-based synthetic weather transfer by +12.3 mAP₅₀, +8.9 mAP_{50:95}, and +7.2 MBIoU on average, confirming stronger synthetic-to-real transfer under challenging-weather domain shifts.

Our work makes three main contributions:

- We propose MsGA-Edit, a training-free two-stage diffusion framework for layout-preserving weather-effect transfer in driving scenes, using classifier-free guidance to adjust visual weather strength while maintaining stable scene geometry.
- We introduce *Structure-Aware Guidance (SAG)* and refine *Multi-source Guided Attention (MsGA)*: SAG extracts semantic, edge, and spatial-feature structural priors through synchronized inversion, while MsGA injects them into self-attention to reduce weather-induced layout corruption.

- We build RoadWeather-34K and demonstrate strong synthetic-to-real benefits across two downstream protocols: fine-tuning detectors with our generated data improves zero-shot DAWN performance by up to +19.5 mAP₅₀ and +22.1 mean matched-box IoU (MBIoU), while on ACDC, MsGA-Edit reaches 77.9% of the ACDC-real mAP₅₀ upper bound.

II. RELATED WORK

A. Weather-Effect Image Synthesis and Image-to-Image Translation

Challenging weather (heavy rain, dense fog, snow) degrades image quality and visibility, harming detection and segmentation performance [22]. Because large, well-annotated datasets in these conditions are costly and scarce, many works synthesize or remove weather effects to expand training data and improve robustness [16]. Early physical-prior methods (e.g., *dark-channel prior*) and analytic degradation models struggle to generalize to diverse or mixed weather [23], [24]. GAN-based translation quickly became the dominant end-to-end approach. Adversarial learning produces sharp, photo-realistic images, and conditional/cycle formulations support both paired and unpaired supervision. Modular generators and discriminators make it easy to add attention, weather cues, and multi-domain conditioning. These advantages led to many variants for weather synthesis; *CycleGAN* established unpaired I2I feasibility [25]. In parallel, non-GAN routes have also been explored, including VAE-style generative models and physics/graphics-based simulators, and more recently diffusion models that enable training-free/zero-shot editing for weather transfer [9]. Representative milestones include *WeatherGAN* for multi-category translation with attention to weather cues [8], asymmetric designs such as *AU-GAN* for rain-night→day [26], and multi-domain conditional translation demonstrating utility for downstream segmentation [27]. However, GAN pipelines often incur a high training cost, can distort safety-critical structures, and offer limited editing-strength adjustment, with synthetic–real gaps persisting [28].

B. Training-based and Few-shot Diffusion Editors

The limits of purely GAN-based translation (training cost, structure distortion, weak editing-strength adjustment) and the brittleness of analytic priors motivate a middle ground between simulators/physics models and the fully training-free diffusion editors discussed next. In this space, *training-based* or *few-shot* diffusion methods enhance controllability and identity fidelity with only lightweight updates. Adapter-style approaches (e.g., *ControlNet*, *T2I-Adapter*, *IP-Adapter*) freeze the base model and learn compact modules to inject spatial or image-level conditions—edges, depth, segmentation, or image prompts—thus enabling composable, layout-aware controls [29]–[31]. In parallel, *personalization* methods fit a concept or subject from a handful of exemplars: *DreamBooth* binds a rare token via few-shot fine-tuning with prior preservation [32]; *Textual Inversion* learns concept embeddings without touching backbone weights [33]; *Custom Diffusion* and *Perfusion* further cut memory/compute by editing cross-attention

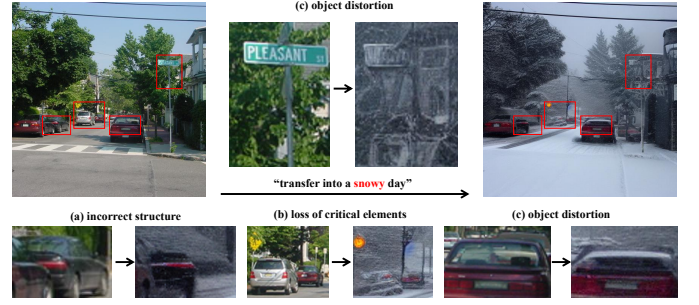


Fig. 3. **Stronger CFG-guided weather prompts can corrupt street-scene layout and semantics.** As the apparent weather effect becomes stronger, *cross-attention leakage* may corrupt geometric layout and semantic structure. Left: the clear-weather source image. Right: the edited snowy result. Red boxes mark three typical failures: (a) incorrect structure, (b) loss of critical elements, and (c) object distortion. Best viewed on screen with zoom.

or rank-one factors, improving multi-concept composition and identity retention [34], [35]. For real-image editing, *Imagic* jointly optimizes a text embedding and a small set of weights to reconstruct and then manipulate a single input with non-rigid changes [36]. For weather-effect transfer, these routes calibrate visual style strength and domain priors from a small curated set of weather exemplars, while adapter-based spatial conditioning helps preserve street-layout geometry; they also complement instruction-tuned editors (e.g., [11]) when task-specific alignment is desired. Taken together, this family of methods forms a pragmatic bridge from data-hungry, task-specific GAN training to the general-purpose, zero-shot diffusion editors we review next.

C. Zero-Shot Editing and Style Transfer with Diffusion Models

Diffusion models [9] have become the dominant backbone for high-fidelity synthesis. *Latent diffusion* [10] performs denoising in a compact VAE latent, enabling efficient, text-conditioned generation at scale; large systems such as *Imagen* [37] and *DALL-E 2* [38] further couple language–vision priors and CLIP-aligned latents to unlock broadly capable, zero-shot editing without task-specific finetuning. In practice, classifier-free guidance and token-aligned cross-attention let text embeddings steer the noise-to-image trajectory while the latent autoencoder concentrates compute on perceptually relevant manifolds. The result is a flexible editing substrate: fine-grained semantics are injected through cross-attention, while global style and color are adjusted by the denoiser’s score field. A minor drawback is that text-only steering can overfit to language priors—when the prompt poorly localizes objects or implies implausible textures, geometry may drift even without finetuning.

On top of these backbones, training-free editors can be organized by how they *constrain content while steering style*. *Attention-aligned* methods explicitly reuse or manipulate cross-attention so the prompt alters appearance but preserves layout. *Pix2Pix-Zero* [39] computes a CLIP-consistent *direction* between source and target prompts and injects this direction during denoising, while *replaying the source cross-attention maps* at each step to anchor spatial correspondences; concretely, keys/values from the source prompt guide where edits apply, and the CLIP direction adjusts how strongly

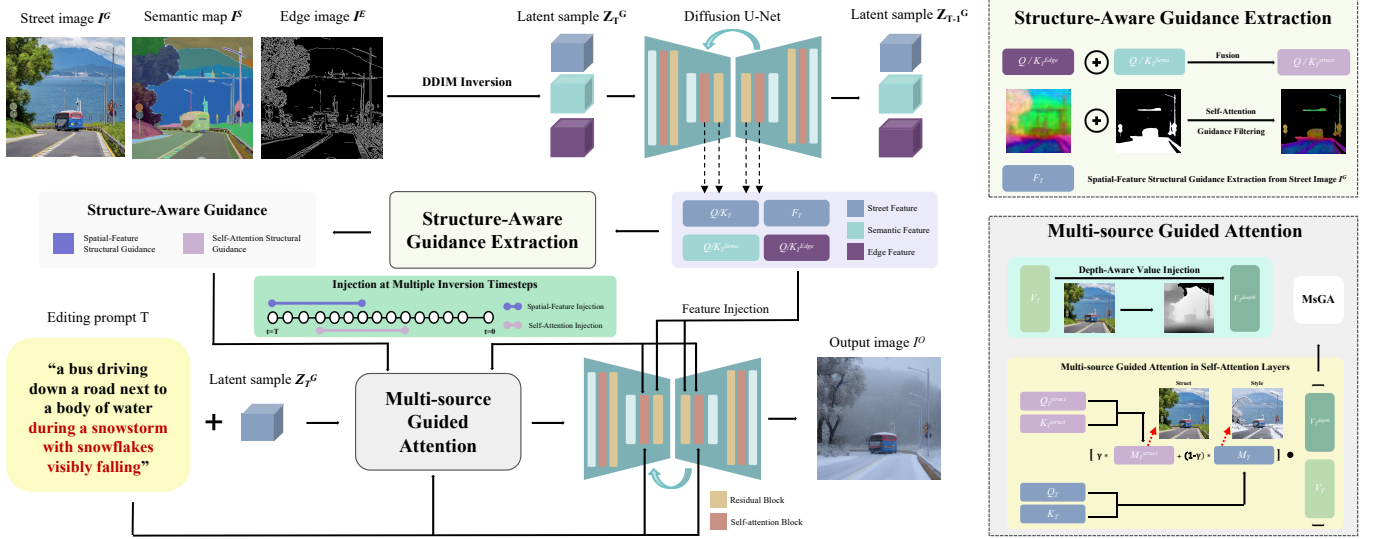


Fig. 4. **Overall framework of MsGA-Edit.** Given an original street image I^G , its semantic map I^S , Canny edge map I^E , and depth map D , MsGA-Edit first performs synchronized DDIM inversion to extract Structure-Aware Guidance. The guidance contains two complementary parts: Self-Attention Structural Guidance, obtained by fusing semantic/edge Q/K carriers and filtering them within the structural mask, and Spatial-Feature Structural Guidance, extracted from the original street-image feature F^G . During Stage 2 DDIM denoising, the Q/K guidance is injected within the DDIM sampling-step window T_{QK}^{inj} to form structure-guided self-attention and activate DVI, while the spatial feature F^G is injected within the DDIM sampling-step window T_F^{inj} to anchor source layout and local appearance. The fixed mask-gated fusion $W_\ell(x) = \gamma M_\ell(x)$ restricts structural intervention to selected regions and keeps non-target regions on the source-preserving feature branch.

features move toward the target description. This typically safeguards coarse geometry but can under-edit fine, texture-level details if the CLIP direction is weak for small objects. *Prompt-to-Prompt* [40] instead *directly edits attention*: it swaps, fuses, or reweights token-level attention maps across selected layers and timesteps, enabling (i) token replacement for localized edits (e.g., “sky, $\rightarrow$, stormy sky”), (ii) mask-like control by freezing maps for tokens that should remain unchanged, and (iii) step-wise schedules to confine edits to early (layout) or late (appearance) denoising phases. The method gives precise, prompt-native control but relies on good token–region alignment; vague prompts or overlapping concepts can cause leakage across parts. Recent analysis [17] delves into the semantics of attention mechanisms in Stable Diffusion, revealing that cross-attention maps often embed object attribution information alongside token weights, leading to editing failures via unintended semantic leakage from original content. In contrast, self-attention maps are pivotal for preserving geometric and shape details during transformations. Leveraging these insights, it proposes a simplified tuning-free procedure that modifies only self-attention maps in selected layers, achieving more stable and efficient edits compared to prior methods. Moving beyond attention reuse, *Plug-and-Play Diffusion Features* [5] injects intermediate *features*—both self-attention and spatial activations—extracted from a guidance image into the target generation trajectory. Practically, the method caches multi-scale features from a guidance run and, within a chosen temporal window, blends them into mid/high-resolution U-Net blocks of the target run, thereby *anchoring semantic layout* while allowing the text to drive style. Feature choice (self-attention early for long-range geometry, spatial activations late for edges/details) and a gentle injection schedule help balance structure and style; however, excessive feature

strength can oversuppress the target style, while weak injection may not fully prevent content drift.

A complementary line first *inverts* the input into a faithful diffusion trajectory and then constrains edits around that path. *LEDITS++* [14] adopts a parameter-free (e.g., DDIM-like) inversion to reconstruct the source image, then applies an *editing schedule* that progressively increases target conditioning while keeping source-aligned signals strong early on. It further derives *implicit masks* from attention magnitude and gradient cues to confine edits to likely regions, so changes remain local and reconstruction fidelity stays high; the limitation is that attention-derived masks can be coarse for tiny structures or crowded scenes. *TF-ICON* [41] improves inversion fidelity by tightening the forward–reverse consistency of trajectories and *composes self-attention* from source and target prompts—e.g., mixing keys/values or blending heads—to obtain smoother cross-domain composition under text guidance. This reduces discontinuities at object boundaries but introduces extra hyperparameters for mixing ratios and may attenuate very strong style shifts. For street-scene stylization specifically, *coupling/blending* strategies aim to keep content while shifting weather. *CycleDiffusion* [15] coordinates paired forward–reverse trajectories, often with a fixed seed and cycle-consistency constraints, so the target path remains *coupled* to the source; this curbs content drift and preserves layout, though the tight coupling can limit aggressive appearance changes or require seed control for reproducibility. *Z-STAR+* [42] blends latent and style features under attention control—e.g., gating cross-attention for structure tokens while injecting style channels in later layers—achieving zero-shot transfer with stronger content anchoring and fewer artifacts. Its main trade-off is tuning: over-blending risks ghosting or desaturation, while under-blending yields weak weather effects. Together, these families

illustrate the central design choice of training-free diffusion editors: either *lock down content* via attention/features and then apply style, or *lock down the trajectory* via inversion and edit around it—each bringing complementary control knobs and failure modes.

D. Synthetic Data for Downstream Perception Tasks

Synthetic data generation is increasingly evaluated by downstream perception utility rather than by visual realism alone. In autonomous driving and challenging-weather perception, generated samples are useful only when they preserve object locations, road topology, and annotation consistency after appearance transfer. Prior weather-transfer work has therefore used synthetic challenging-weather images to improve segmentation or detection robustness [27], [28], but the synthetic–real gap remains sensitive to layout distortion and artifact-induced label noise. Beyond driving scenes, recent task-oriented generators show the same principle in other perception domains. AeroGen proposes a layout-controllable diffusion model for remote-sensing object detection and uses generated samples to improve detectors on DIOR, DIOR-R, and HRSC [43]. MFA-CycleGAN synthesizes sonar images to augment underwater object detection for AUVs, showing that domain-specific image generation can improve detector training when real data are limited [44]. These studies motivate our downstream evaluation protocol: synthetic weather images should be assessed not only by image-level fidelity, but also by whether detectors trained on them transfer to real challenging-weather benchmarks.

III. PRELIMINARIES

Diffusion models learn a data distribution by reversing a fixed Gaussian noising process. In DDPM [45], the forward (fixed) marginal is

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}). \quad (1)$$

Training minimizes a simple noise-prediction loss

$$\mathcal{L} = \mathbb{E}_{x_0, t, \epsilon} \|\epsilon - \epsilon_\theta(x_t, t)\|_2^2, \quad (2)$$

and generation iteratively denoises from $x_T \sim \mathcal{N}(0, \mathbf{I})$.

To accelerate sampling, DDIM [9] provides a non-Markovian (optionally deterministic) update:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{x}_0(x_t, t) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \epsilon_\theta(x_t, t) + \sigma_t \epsilon, \quad (3)$$

where $\hat{x}_0(x_t, t) = (x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t)) / \sqrt{\bar{\alpha}_t}$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, and $\sigma_t = \eta \sqrt{(1 - \bar{\alpha}_{t-1}) / (1 - \bar{\alpha}_t)} \sqrt{1 - \bar{\alpha}_t / \bar{\alpha}_{t-1}}$ with $\eta \in [0, 1]$. The deterministic DDIM case corresponds to $\eta = 0$.

Latent diffusion (LDM) moves diffusion from pixel space to a VAE latent space z to cut compute while preserving semantics [10]. An encoder E maps $x \rightarrow z$, diffusion/denoising proceeds in z , and a decoder D reconstructs $x \approx D(z)$. Text/image conditions y are injected via cross-attention inside the U-Net denoiser, realizing a conditional transition $p_\theta(z_{t-1} | z_t, t, y)$ that supports high-resolution synthesis and precise, edit-friendly control.

In this work we adopt *Stable Diffusion v1.4*, an *LDM* comprising a pretrained VAE, a $\sim 860\text{M}$ -parameter U-Net denoiser, and a CLIP text encoder, pretrained on LAION subsets; it supports 512×512 generation, text guidance, and image-to-image editing [10], [38].

IV. METHOD

A. Overview

We present MsGA-Edit, a training-free diffusion editing framework for layout-preserving weather-effect transfer in driving scenes. As shown in Fig. 4, the framework contains two stages. The first stage, *Structure-Aware Guidance Extraction*, runs synchronized DDIM inversion on the original street image I^G , the semantic map I^S , and the Canny edge map I^E . It extracts two structural priors from the aligned inversion trajectory: Self-Attention Structural Guidance from semantic/edge Q/K carriers, and Spatial-Feature Structural Guidance from the original street-image feature $F_{\ell, t}^G$. The second stage, *Multi-source Guided Attention* (MsGA), performs target-weather generation with DDIM denoising and injects the two cached structural priors into selected self-attention layers. The filtered Q/K guidance is used to build the structure-guided self-attention branch, while the spatial-feature guidance anchors the target trajectory to the source street-image feature branch. DVI is coupled with the Q/K-guided MsGA branch: whenever the Q/K branch is active, DVI modulates the weather value tensor according to scene depth before the mask-gated fusion.

The injection design follows the “Injection at Multiple DDIM Sampling Steps” module in Fig. 4, and all guidance injection operations are applied in Stage 2. We use $N_{\text{DDIM}} = 50$ DDIM sampling steps and denote the sampling-step index as $s \in \{1, \dots, N_{\text{DDIM}}\}$, where $s = 1$ is the initial high-noise denoising stage and $s = N_{\text{DDIM}}$ is the final low-noise stage. Each sampling step s corresponds to a cached DDIM-inversion timestep t_s in the original diffusion noise schedule. Therefore, we define the Stage 2 guidance windows on the DDIM sampling-step index as

$$T_{\text{QK}}^{\text{inj}} \subseteq \{1, \dots, N_{\text{DDIM}}\}, \quad T_{\text{F}}^{\text{inj}} \subseteq \{1, \dots, N_{\text{DDIM}}\}. \quad (4)$$

Here $T_{\text{QK}}^{\text{inj}}$ controls the injection of filtered Q/K carriers for MsGA self-attention and simultaneously activates DVI, while $T_{\text{F}}^{\text{inj}}$ controls the injection of the cached street-image feature F_{ℓ, t_s}^G . Whenever s falls within a guidance window, the corresponding cached carriers at $t = t_s$ are retrieved from the synchronized inversion trajectory. In figures and ablations, early, middle, and late injection respectively refer to low, intermediate, and high values of the DDIM sampling index s .

In this design, the semantic and edge branches provide complementary self-attention carriers. The semantic branch supplies category-level correspondences among roads, sidewalks, buildings, vehicles, pedestrians, and traffic infrastructure. The edge branch supplies boundary-level geometry for thin structures, object contours, lane markings, poles, and building outlines. The original street-image branch supplies $F_{\ell, t}^G$ to anchor the target trajectory to source layout and local

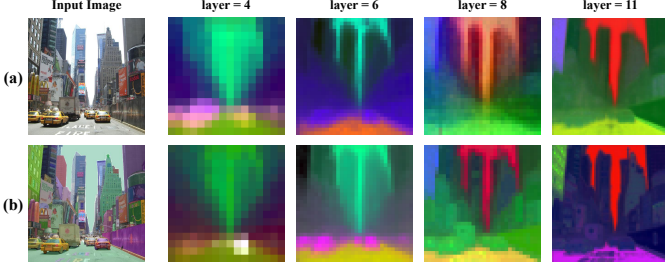


Fig. 5. **Self-attention visualization with PCA.** (a) Input street image and (b) its semantic map. Self-attention PCA visualizations for layers 4, 6, 8, and 11. The principal components reveal enhanced structural clustering: similar urban elements share consistent colors across the semantic layout. Notably, the semantic map provides richer structural information, enabling more accurate clustering of street components like roads and buildings in the attention space.

appearance. The main editing parameters are λ for semantic–edge Q/K fusion, M_ℓ for structural-token filtering, fixed γ for mask-gated fusion, $T_{\text{QK}}^{\text{inj}}$ and $T_{\text{F}}^{\text{inj}}$ for timestep-selective injection, and β and s_{DVI} for DVI. All denoising-network, VAE, segmentation, edge, and depth-estimation modules are frozen.

B. Stage 1: Structure-Aware Guidance Extraction

Given an original clear-weather street image I^G , we construct a semantic map I^S and a Canny edge map I^E . The semantic map is predicted by a frozen SegFormer-B5 model pretrained on Cityscapes [46] and converted into a 3-channel color-coded label image with a fixed palette. The edge map is obtained by the Canny detector [47] and replicated to three channels. All three inputs are resized and normalized to match the latent-diffusion backbone. For each modality $m \in \{G, S, E\}$, the VAE encoder produces

$$z_0^m = E(I^m). \quad (5)$$

We then apply deterministic DDIM inversion under an empty prompt with a shared timestep schedule, yielding synchronized trajectories $\{z_t^m\}_{t=0}^{T_{\text{max}}}$. The shared frozen denoiser and the shared inversion schedule align semantic/edge Q/K carriers and street-image features at the same layer ℓ and timestep t .

Following the feature-caching strategy of Plug-and-Play Diffusion [5], we cache intermediate self-attention carriers and spatial features from selected U-Net layers $\ell \in \mathcal{L}$. The semantic and edge branches provide Q/K carriers for Self-Attention Structural Guidance, and the original street-image branch provides $F_{\ell,t}^G$ for Spatial-Feature Structural Guidance. The cached components are stored for their corresponding Stage 2 guidance windows:

$$\begin{aligned} (Q_{\ell,t_s}^S, K_{\ell,t_s}^S), (Q_{\ell,t_s}^E, K_{\ell,t_s}^E), \quad & s \in T_{\text{QK}}^{\text{inj}}, \\ F_{\ell,t_s}^G, \quad & s \in T_{\text{F}}^{\text{inj}}. \end{aligned} \quad (6)$$

Here $(Q_{\ell,t_s}^S, K_{\ell,t_s}^S)$ encodes semantic-region layout, $(Q_{\ell,t_s}^E, K_{\ell,t_s}^E)$ encodes edge-level geometry, and F_{ℓ,t_s}^G preserves source spatial details. Fig. 5 shows that semantic-map attention forms coherent structural clusters for roads, buildings, sidewalks, and vehicles, while edge cues complement it with sharp boundaries and thin structures.

1) *Self-Attention Structural Guidance Fusion*: The first operation in Structure-Aware Guidance Extraction is Q/K fusion. For each selected layer and timestep, we fuse semantic and edge carriers as

$$\begin{aligned} Q_{\ell,t}^{\text{fuse}} &= \lambda Q_{\ell,t}^S + (1 - \lambda) Q_{\ell,t}^E, \\ K_{\ell,t}^{\text{fuse}} &= \lambda K_{\ell,t}^S + (1 - \lambda) K_{\ell,t}^E. \end{aligned} \quad (7)$$

The coefficient $\lambda \in [0, 1]$ controls the balance between semantic-region structure and boundary-level structure. A larger λ emphasizes semantic layout, and a smaller λ emphasizes Canny contours. For single-source variants, we set $\lambda = 1$ for semantic-only guidance and $\lambda = 0$ for edge-only guidance.

2) *Self-Attention Guidance Filtering*: We construct a binary structural mask $M \in \{0, 1\}^{H \times W}$ from semantic labels. The mask covers structure-critical regions, including roads, road markings, sidewalks, vehicles, pedestrians, poles, traffic signs, and buildings. For layer ℓ , the mask is resized to the corresponding token grid:

$$M_\ell = \Pi_\ell(M), \quad M_\ell(x) \in [0, 1], \quad (8)$$

where $\Pi_\ell(\cdot)$ denotes downsampling to the token resolution of layer ℓ . The mask filters the fused carriers and retains the tokens in the selected structural region:

$$\begin{aligned} \tilde{Q}_{\ell,t}^{\text{str}} &= M_\ell \odot Q_{\ell,t}^{\text{fuse}}, \\ \tilde{K}_{\ell,t}^{\text{str}} &= M_\ell \odot K_{\ell,t}^{\text{fuse}}. \end{aligned} \quad (9)$$

Here M_ℓ is broadcast along the channel dimension. The retained pair $(\tilde{Q}_{\ell,t}^{\text{str}}, \tilde{K}_{\ell,t}^{\text{str}})$ is used as Self-Attention Structural Guidance in Stage 2; at DDIM step s , the target denoising branch retrieves the cached pair with $t = t_s$. The filtering step localizes semantic/edge guidance to structure-critical tokens and limits changes in unrelated regions.

3) *Spatial-Feature Structural Guidance Extraction from Street Image*: The original street-image branch caches F_{ℓ,t_s}^G at the scheduled inversion timesteps corresponding to $s \in T_{\text{F}}^{\text{inj}}$. This feature is used as Spatial-Feature Structural Guidance and carries source layout, local appearance, and fine spatial details. The output of Structure-Aware Guidance Extraction therefore consists of the filtered Q/K pair in Eq. (9) and the cached street-image feature F_{ℓ,t_s}^G in Eq. (6). A monocular depth map $D(x)$ is also estimated from I^G by MiDaS [48] and used by DVI in Stage 2.

Fig. 6 illustrates the complementary roles of the two guidance branches. Spatial-Feature Structural Guidance retains the original street-image feature distribution and stabilizes coarse layout and local appearance. Self-Attention Structural Guidance reuses mask-filtered semantic/edge Q/K carriers to recover semantic correspondences among structure-critical regions and assign object-level details to the correct spatial locations. Together, the spatial-feature branch preserves the locations of roads, vehicles, and buildings, and the self-attention branch refines the semantic details at those locations.

C. Stage 2: MsGA with DVI and Mask-Gated Fusion

Target weather generation starts from the inverted street-image latent $z_{T_{\text{max}}}^G$ and the editing prompt c_w . This second



Fig. 6. **Complementary effects of Self-Attention Structural Guidance and Spatial-Feature Structural Guidance.** Both guidance branches contain source street-scene information but act on different representations. Spatial-Feature Structural Guidance anchors the original feature layout and local appearance, while Self-Attention Structural Guidance restores semantic/edge-aware token correspondences. Using only one branch may preserve partial structure but still leaves layout drift or semantic-detail loss; using both branches provides stronger protection for road topology, vehicles, buildings, and thin urban structures.

stage uses DDIM denoising with N_{DDIM} sampling steps. We denote the DDIM sampling index by $s \in \{1, \dots, N_{\text{DDIM}}\}$, where $s = 1$ is the initial high-noise denoising step and $s = N_{\text{DDIM}}$ is the final low-noise step. Each DDIM step s corresponds to an original diffusion timestep t_s on the underlying noise schedule. At each selected self-attention layer, MsGA injects the cached structural priors according to two Stage 2 guidance windows: $T_{\text{QK}}^{\text{inj}}$ for Self-Attention Structural Guidance and $T_{\text{F}}^{\text{inj}}$ for Spatial-Feature Structural Guidance. The Q/K window activates structure-guided self-attention, and the feature window activates source spatial-feature injection. DVI is not assigned an independent denoising window; it is coupled with the Q/K-based MsGA branch and is applied whenever $s \in T_{\text{QK}}^{\text{inj}}$.

1) *Depth-Aware Value Injection:* DVI models distance-dependent weather accumulation before attention-output fusion. Given the depth map $D(x)$, we define the depth valve as

$$U_D(x) = 1 - \exp\left(-\beta \frac{D(x)}{255}\right), \quad (10)$$

where β controls the sensitivity of weather accumulation to depth. The normalized depth valve is

$$\bar{U}_D(x) = s_{\text{DVI}} \mathcal{N}(U_D(x)) + (1 - s_{\text{DVI}}), \quad (11)$$

where $\mathcal{N}(\cdot)$ denotes min-max normalization and $s_{\text{DVI}} \in [0, 1]$ controls the global strength of DVI modulation. For each selected layer, $\bar{U}_D$ is resized to the token grid and broadcast over channels. The weather value is selected as

$$V_{\ell, t_s}^{\text{d}} = \begin{cases} V_{\ell, t_s}^{\text{w}} \odot \Pi_{\ell}(\bar{U}_D), & s \in T_{\text{QK}}^{\text{inj}}, \\ V_{\ell, t_s}^{\text{w}}, & s \notin T_{\text{QK}}^{\text{inj}}. \end{cases} \quad (12)$$

DVI controls the depth distribution of weather appearance inside the MsGA Q/K injection window, while the following MsGA step controls structural injection and spatial-feature preservation.

2) *Timestep-Selective MsGA and Spatial-Feature Injection:* For $s \in T_{\text{QK}}^{\text{inj}}$, MsGA computes a structure-guided self-attention map from the filtered Q/K carriers:

$$A_{\ell, t_s}^{\text{str}} = \text{softmax}\left(\frac{\tilde{Q}_{\ell, t_s}^{\text{str}} (\tilde{K}_{\ell, t_s}^{\text{str}})^{\top}}{\sqrt{d_k}}\right). \quad (13)$$

The source-preserving branch is selected by the feature-injection window:

$$\hat{F}_{\ell, t_s} = \begin{cases} F_{\ell, t_s}^{\text{G}}, & s \in T_{\text{F}}^{\text{inj}}, \\ F_{\ell, t_s}^{\text{tar}}, & s \notin T_{\text{F}}^{\text{inj}}, \end{cases} \quad (14)$$

where $F_{\ell, t_s}^{\text{tar}}$ denotes the current target spatial feature at layer ℓ and DDIM step s (with original diffusion timestep t_s). The fixed spatial fusion weight is

$$W_{\ell}(x) = \gamma M_{\ell}(x), \quad \gamma \in [0, 1]. \quad (15)$$

The parameter γ is fixed during denoising, which makes the strength of mask-gated fusion independent of the timestep-window choice. For $s \in T_{\text{QK}}^{\text{inj}}$, the MsGA output is

$$O_{\ell, t_s} = W_{\ell} \odot (A_{\ell, t_s}^{\text{str}} V_{\ell, t_s}^{\text{d}}) + (1 - W_{\ell}) \odot \hat{F}_{\ell, t_s}. \quad (16)$$

Eq. (16) corresponds to the Multi-source Guided Attention

block in Fig. 4. We emphasize that MsGA is a mask-gated weighted fusion mechanism for layout preservation, not a strict factorization of weather style and geometric structure. The weather appearance is still driven by the target prompt and CFG denoising path, while the structural priors bias the self-attention branch and the source-feature branch to reduce layout drift. Inside the structural mask, the filtered semantic-edge Q/K carriers determine the self-attention correspondence and V_{ℓ, t_s}^{d} provides the depth-aware weather value. Outside the mask, $M_{\ell}(x) = 0$, so the layer follows the spatial-feature branch $\hat{F}_{\ell, t_s}$. For $s \notin T_{\text{QK}}^{\text{inj}}$, the target U-Net keeps its original self-attention computation. If $s \in T_{\text{F}}^{\text{inj}}$, F_{ℓ, t_s}^{G} is still injected through the spatial-feature branch; if $s \notin T_{\text{QK}}^{\text{inj}} \cup T_{\text{F}}^{\text{inj}}$, the layer follows the original target denoising branch. Thus $T_{\text{QK}}^{\text{inj}}$ and $T_{\text{F}}^{\text{inj}}$ separately control self-attention correspondence and source-layout anchoring.

3) *Classifier-Free Weather Guidance Scale:* The apparent strength of the target weather style is adjusted by the classifier-free guidance scale ω . Following [5], the denoising prediction is

$$\epsilon_{\theta}(x_t, t, c) = \epsilon_{\theta}(x_t, t, \emptyset) + \omega \Delta \epsilon_{\theta}(x_t, t, c_w), \quad (17)$$

where

$$\Delta \epsilon_{\theta}(x_t, t, c_w) = \epsilon_{\theta}(x_t, t, c_w) - \epsilon_{\theta}(x_t, t, \emptyset). \quad (18)$$

Here c_w denotes the target weather prompt. The scale ω is an editing knob for the visual strength of the requested weather appearance. It is not calibrated to physical meteorological variables such as rainfall rate, particle size, or fog



Fig. 7. **CFG-guided weather appearance spectrum generated by MsGA-Edit.** The figure shows synthesized weather/appearance variations across four target styles. Each column increases the classifier-free guidance scale to strengthen the visual weather appearance while preserving the underlying scene structure.

visibility. Accordingly, our method is evaluated as a layout-preserving weather-effect editor under stronger CFG-driven weather prompts. The structural parameters are λ , M_ℓ , fixed γ , $T_{\text{QK}}^{\text{inj}}$, and $T_{\text{F}}^{\text{inj}}$. The depth-related parameters are β and s_{DVI} , which shape relative depth-dependent appearance inside the MsGA Q/K window $T_{\text{QK}}^{\text{inj}}$. In implementation, MsGA is applied only to selected self-attention layers $\mathcal{L}$.

V. EXPERIMENTS

A. CFG-Guided Weather Appearance Spectrum Generation

To construct a dense spectrum of weather appearances for each scene, we generate progressively strengthened variants of every target weather/appearance style in a fully automated manner. In this setting, all structural extraction is completed before editing, and the actual guidance injection is performed only in the second-stage DDIM denoising process. The structural parameters λ and γ are kept fixed during denoising: λ determines the semantic–edge balance in Self-Attention Structural Guidance, and γ determines the fixed mask-gated fusion strength in $W_\ell(x) = \gamma M_\ell(x)$. The apparent weather strength is adjusted through the classifier-free guidance scale ω ; we do not claim that these grades correspond to measured rainfall rate, fog visibility, or other meteorological quantities. Specifically, ω controls how strongly the target weather prompt affects the denoising prediction under classifier-free guidance. The DVI strength s_{DVI} only regulates the relative depth weighting of the weather value tensor inside the MsGA branch: higher values emphasize far-field tokens and can produce stronger distance-dependent visual degradation, such as denser far-field fog, while keeping near-field objects comparatively clearer. The DVI sensitivity β is kept as the depth-response sharpness parameter.

The default configuration uses $N_{\text{DDIM}} = 50$ DDIM sampling steps in Stage 2, $\lambda = 0.67$, fixed $\gamma = 0.15$, DDIM stochasticity $\eta = 0.1$, DVI depth sensitivity $\beta = 0.8$, and DVI strength $s_{\text{DVI}} = 0.2$. Stage 1 only caches Structure-Aware Guidance, while Stage 2 applies the two guidance

injections during target-weather DDIM denoising. We report the active windows using the DDIM sampling-step index $s \in \{1, \dots, 50\}$; the cached guidance at each active step is retrieved from the corresponding inversion timestep t_s . In our default setting, $T_{\text{QK}}^{\text{inj}} = [1, 35]$ is used for Self-Attention Structural Guidance, namely the MsGA Q/K injection, and $T_{\text{F}}^{\text{inj}} = [1, 25]$ is used for Spatial-Feature Structural Guidance, namely the source spatial-feature injection.

By sweeping ω over a small, fixed set of levels while keeping λ , γ , $T_{\text{QK}}^{\text{inj}}$, $T_{\text{F}}^{\text{inj}}$, and DVI parameters unchanged, MsGA-Edit yields six distinct CFG-guided appearance grades for each weather/appearance style without model retraining or prompt modification.

An example progression is shown in Fig. 7, where visual weather appearance becomes stronger while global geometry and semantic consistency remain stable.

B. RoadWeather-34K Dataset

We introduce **RoadWeather-34K**, a large-scale synthetic benchmark for layout-preserving weather transfer in driving scenes. RoadWeather-34K is built upon clear-weather street images strictly and manually curated from three public driving corpora—KITTI [2], nuScenes [3], and Cityscapes [4]. To ensure high-quality structural priors for weather transfer, we retain 960 images with information-rich street content and clean geometry cues: clear drivable areas and lane markings, visible sidewalks/curbs, and diverse semantic elements such as vehicles, pedestrians, buildings, and traffic infrastructure (e.g., lights and signs). We further discard samples with weak road context (e.g., empty scenes), severe occlusion, low visibility, extreme exposure, or prominent motion blur. Each selected image is center-cropped and resized to 512×512 and stored in PNG format.

Using a frozen Stable Diffusion v1.4 latent backbone with DDIM sampling [9], [10], our MsGA-Edit framework synthesizes six target weather/appearance styles—snowy, rainy, foggy, sandstorm, autumn, and dawn—for each source image. For every style, we generate six CFG-guided visual appearance variants, producing a dense spectrum of 36 weather/appearance configurations per scene without any fine-tuning. We apply strict dataset construction rules to maintain structural fidelity and usability for perception: generated samples that exhibit obvious layout corruption, semantic drift, or disruptive artifacts are excluded. Overall, this pipeline yields 34,560 stylized images in total; importantly, the 34K in RoadWeather-34K refers to the total number of images (across all scenes, weather/appearance styles, and appearance grades). The resulting benchmark spans a wide range of visual domain shifts and degradation appearances, and is tailored for robust perception and downstream evaluation under challenging weather.

C. Comparative Evaluation and Metrics

We compare MsGA-Edit against five representative diffusion-based editing and zero-shot image translation methods: *PnP* [5] (our main baseline), *Instruct-Pix2Pix* [11], *DiffuseIT* [12], *Edit Friendly DDPM Inversion* [13], and *LED-ITS++* [14]. All methods operate on the same 512×512 inputs,



Fig. 8. **Qualitative comparison with diffusion-based weather-editing baselines.** We compare MsGA-Edit with Instruct-Pix2Pix, DiffuseIT, Edit Friendly DDPM Inversion, LEDITS++, and PnP under identical target-weather prompts covering snowstorm, sandstorm, heavy rain, fog, night, and frost. Compared with the baselines, MsGA-Edit better preserves road layout, vehicles, and background structures while producing the requested weather appearance.

use identical prompts, and adopt the same 50 DDIM sampling steps.

To jointly assess weather editing effects and content preservation, we report multiple complementary metrics. Paired ground-truth targets are not available for this editing task. Unless otherwise specified, all image-level preservation metrics are computed *with respect to the original clear-weather input* (i.e., the pre-edit source image). Specifically, pixel-level fidelity is measured by mean squared error (MSE) and peak signal-to-noise ratio (PSNR), structural similarity is quantified with SSIM, and perceptual distance is evaluated with LPIPS [49]. In addition, we employ CLIP-based similarities [50]: *CLIP-img* computes the cosine similarity between the CLIP image embeddings of the generated image and the source image, reflecting scene/layout preservation in the embedding space; *CLIP-word* computes the cosine similarity between the CLIP image embedding of the generated image and the CLIP text embedding of the weather prompt used for guidance, measuring text-image semantic alignment. Quantitative comparisons are summarized in Table I, and qualitative examples are shown in Fig. 8.

DiffuseIT tends to introduce larger unintended deviations from the source scene, leading to degraded preservation scores (e.g., lower CLIP-img and SSIM) and weaker prompt alignment (lower CLIP-word). *Instruct-Pix2Pix* attains strong text-image alignment (high CLIP-word) and relatively high embedding-level preservation (high CLIP-img), yet it often overlays weather-like textures without consistent coupling to scene geometry, which harms pixel-level and perceptual fidelity. *Edit Friendly DDPM Inversion* produces visually plausible weather particles with favorable perceptual scores, but it can soften small structures and reduce layout consistency (lower CLIP-img). *LEDITS++* best preserves structure and naturalness according to PSNR/SSIM and LPIPS, and achieves high CLIP-img, yet it may under-express the requested weather semantics in driving scenes, resulting in lower CLIP-word. *PnP* is conservative in preserving low-level appearance (low MSE), but its depth-agnostic design may under-model depth-varying phenomena such as far-field fog and spatially varying snow density. We only benchmark against methods with publicly available implementations. Although *Qwen-Image-Edit* is visualized in Fig. 1 as a representative

TABLE I

QUANTITATIVE COMPARISON. WE COMPARE OUR APPROACH WITH STATE-OF-THE-ART DIFFUSION-BASED IMAGE EDITING METHODS ACROSS MULTIPLE EVALUATION METRICS. TOP-2 SCORES ARE HIGHLIGHTED IN BOLD. $\uparrow$ INDICATES HIGHER IS BETTER AND $\downarrow$ INDICATES LOWER IS BETTER.

Method	CLIP Similarity		MSE $\times 10^4$ $\downarrow$	Image Preservation		
	CLIP-word $\uparrow$	CLIP-img $\uparrow$		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Instruct-Pix2Pix	89.1%	83.5%	156.04	12.179	0.571	0.35
DiffuseIT	69.4%	74.7%	316.17	13.173	0.295	0.43
Edit Friendly DDPM Inv.	90.8%	78.3%	190.42	18.841	0.665	0.27
LEDITS++	85.8%	84.2%	148.19	21.976	0.654	0.29
Plug-and-Play	88.2%	77.9%	142.31	20.743	0.629	0.39
Ours	90.2%	79.2%	141.60	21.956	0.674	0.30

TABLE II

COMPONENT ABLATION OF STRUCTURE-AWARE GUIDANCE, **MsGA**, AND **DVI**. WE REPORT CLIP-WORD SIMILARITY $\uparrow$, PSNR $\uparrow$, AND SSIM $\uparrow$ METRICS.

Method	CLIP-word $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
PnP (baseline)	88.2%	20.743	0.629
+ SAG only	89.1%	21.552	0.671
+ SAG + MsGA (w/o DVI)	89.3%	21.784	0.671
+ SAG + MsGA + DVI	90.2%	21.956	0.674

recent editor, it was not open-sourced at the time of our experiments; therefore, we do not report its results in our quantitative tables.

From this perspective, *MsGA-Edit* exhibits a more balanced trade-off between structure preservation and semantic alignment. It achieves low pixel error and strong SSIM, while maintaining competitive CLIP-word and CLIP-img compared with other baselines. These results suggest that incorporating Structure-Aware Guidance, depth-aware value modulation, and mask-gated multi-source guided attention can help constrain the denoising trajectory for better layout fidelity, while still allowing CFG-adjusted weather appearance injection.

D. Ablation Studies

1) *Effect of Structure Priors, MsGA, and DVI*: We conduct a systematic ablation study to evaluate the individual contributions of Structure-Aware Guidance (SAG), Multi-source Guided Attention (MsGA), and Depth-Aware Value Injection (DVI). Four configurations are compared: (1) *PnP* as the baseline; (2) *SAG only*, which uses Self-Attention Structural Guidance and Spatial-Feature Structural Guidance without the full mask-gated MsGA fusion; (3) *SAG+MsGA* without DVI; and (4) the full model with *SAG+MsGA+DVI*.

Quantitative results in Table II reveal a clear progression of improvements. The baseline PnP method achieves moderate performance across all metrics. Introducing SAG alone improves CLIP-word similarity from 88.2% to 89.1% by enhancing semantic alignment, while boosting PSNR from 20.743 to 21.552 and SSIM from 0.629 to 0.671, indicating effective structural stabilization. The integration of the mask-gated MsGA fusion further improves CLIP-word similarity to 89.3% and PSNR to 21.784 while maintaining SSIM at 0.671. The full model with DVI achieves the optimal balance, delivering the highest CLIP-word similarity (90.2%) and SSIM (0.674) with competitive PSNR (21.956) while preserving strong semantic alignment.

The component progression also explains the visual behavior of the model through the component-level ablation results. The PnP baseline tends to suffer from structural drift and artifacts in complex urban elements. SAG mitigates layout distortion by combining mask-filtered Self-Attention Structural Guidance with source Spatial-Feature Structural Guidance, although it can still oversmooth fine details such as lane markings. The addition of the mask-gated MsGA fusion improves preservation of thin structures and traffic signs by activating the structure-guided self-attention branch only inside M_ℓ . Finally, the full model with DVI produces more visually plausible results by introducing depth-dependent appearance weighting that becomes stronger in far-field regions while protecting near-field structural details from over-stylization.

The depth-aware modulation in DVI effectively addresses the "floating layer" artifact common in uniform weather editing approaches, showing that spatially varying injection based on monocular depth estimation is useful for realistic-looking weather transfer in street scenes, although it is not a calibrated physical scattering model.

2) *Timestep Windows for Structural Guidance Injection*: We further analyze the temporal placement of the two structural guidance branches in Fig. 9. All injections are performed in Stage 2 DDIM denoising, and Stage 1 caches the corresponding guidance signals. The ablation varies the DDIM sampling-step windows T_F^{inj} for Spatial-Feature Structural Guidance and T_{QK}^{inj} for Self-Attention Structural Guidance. The spatial-feature branch is closely tied to coarse layout anchoring. Its window affects whether vehicles, buildings, and road regions remain in the correct positions. The Q/K-based self-attention branch controls semantic attribution and fine details at these positions. Its window affects whether specific objects receive plausible contours, category-consistent textures, and coherent local structures. These results support a clear division of labor: Spatial-Feature Structural Guidance mainly preserves *where* street components are located, and Self-Attention Structural Guidance preserves *what* semantic structures should be formed there.

3) *Where to Inject MsGA: Self- vs. Cross-Attention*: Fig. 10 compares different choices of where to inject MsGA. The baseline PnP method (column 2) generates basic weather effects with visible geometric inaccuracies and missing objects. Cross-attention injection (column 3) strongly couples structural priors with text-driven weather semantics. As a result, the scene layout is drastically altered and essential street information is lost. Cross-attention mediates the interaction

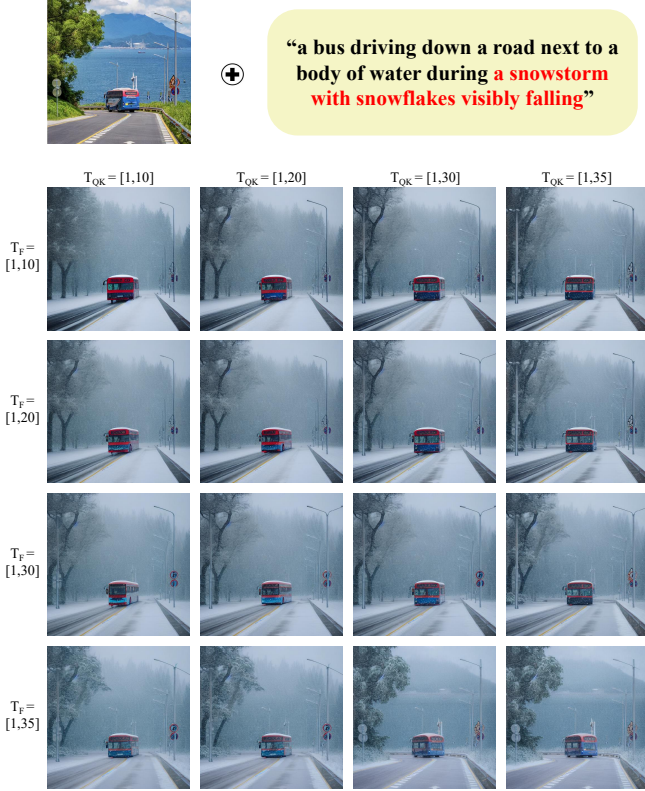


Fig. 9. **Ablation on timestep-selective injection of Spatial-Feature Structural Guidance and Self-Attention Structural Guidance.** The horizontal axis denotes the DDIM sampling-step injection window T_F^{inj} for Spatial-Feature Structural Guidance, and the vertical axis denotes the DDIM sampling-step injection window T_{QK}^{inj} for Self-Attention Structural Guidance. Spatial-feature injection mainly determines coarse scene layout, such as the positions of vehicles, buildings, and road regions. Q/K-based self-attention guidance mainly assigns semantic details and object-specific structure to these locations. The best results are obtained when the two windows are coordinated in Stage 2 DDIM denoising.

between image tokens and text embeddings, so perturbing its queries and keys destabilizes the balance between content preservation and style application.

Applying MsGA exclusively to self-attention layers (column 4) yields the best visual fidelity. Self-attention operates within the image feature space and captures long-range spatial dependencies and intrinsic geometric relationships. Mask-filtered Self-Attention Structural Guidance reinforces these internal correspondences and constrains the denoising process to respect the original scene layout. The text prompt remains responsible for weather appearance. This configuration suppresses style leakage—unwanted distortion of structural elements—and allows depth-aware weather appearance to be integrated.

E. Downstream Perception Enhancement on Real Challenging-Weather Data

Challenging weather remains one of the most persistent bottlenecks for real-world intelligent transportation systems. Rain, fog, snow, and dust reduce visual quality through low contrast, scattering, and motion streaks. They also create a clear domain gap that makes detectors trained on clear-weather data fail in deployment. Collecting and annotating large-scale

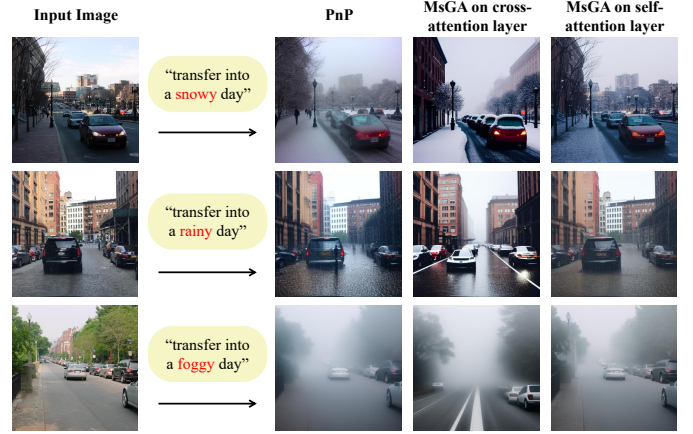


Fig. 10. **Self-attention versus cross-attention injection.** Weather transfer results when applying MsGA to different attention layers. Baseline PnP (column 2) yields plausible snowy/foggy effects but suffers from structural distortions and object loss. Applying MsGA to cross-attention (column 3) severely disrupts the layout and destroys street semantics. Applying MsGA to self-attention (column 4) preserves road topology and critical objects while suppressing style leakage.

real challenging-weather datasets is costly and often unsafe. Recent progress in generative image modeling provides a practical alternative: layout-consistent weather variations can be used as training-time augmentation to improve robustness.

We design two downstream evaluations to verify whether MsGA-Edit produces useful training data for real challenging-weather perception. The first setting follows the DAWN protocol, where detectors are fine-tuned with the curated RoadWeather-34K subset and evaluated on real DAWN images. The second setting uses a stricter synthetic-to-real evaluation on ACDC [51], where all synthetic training images are generated from Cityscapes clear-weather sources and the final evaluation is conducted on real ACDC rain, snow, and fog test images. The quantitative results for both settings are summarized in Fig. 11.

We evaluate on DAWN (Detection in Adverse Weather Nature) [21], a real challenging-weather benchmark composed of 1,000 natural driving scenes (non-rendered, non-synthetic) spanning four weather categories: fog, rain, snow, and sand/dust storms. DAWN images are collected in the wild via web visual search (e.g., Google/Bing) using weather-related keywords, followed by human-in-the-loop filtering to ensure genuine challenging-weather characteristics and diverse driving scenes. Importantly, DAWN provides manual object detection annotations: the authors label bounding boxes using LabelMe, resulting in 7,845 total bounding boxes over six road-agent categories (car, bus, truck, motorcycle, bicycle, and person for pedestrian/cyclist). This makes DAWN a suitable testbed for verifying whether synthetic augmentation can transfer to real challenging-weather perception.

Our downstream goal is to directly measure object recognition/detection robustness under real challenging weather, focusing on three safety-critical classes: car, truck, and person. Concretely, during evaluation on DAWN we retain DAWN ground-truth boxes of these three classes and ignore other categories (e.g., bus/motorcycle/bicycle) to keep the label

(a) Detectors fine-tuned on RoadWeather-34K and evaluated on real DAWN images



(b) Synthetic-to-real effectiveness of generated adverse-weather data on ACDC

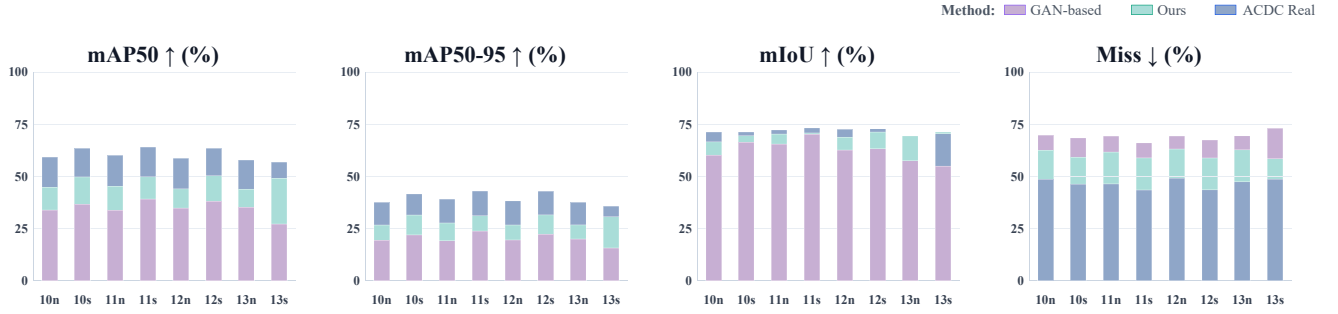


Fig. 11. **Downstream detection results on real challenging-weather benchmarks.** (a) Detectors are fine-tuned on the curated RoadWeather-34K strong-weather subset and evaluated on real DAWN images. Light/dark bars denote the original and fine-tuned detectors. (b) Synthetic-to-real transfer on ACDC: YOLO-series variants are trained from scratch with Cityscapes-derived challenging-weather images generated by GAN-based weather transfer or MsGA-Edit, and are tested on real ACDC rain, snow, and fog images. ACDC-real, trained with real ACDC challenging-weather training images, serves as the real-data upper-bound reference. For Miss, lower values are better.

space consistent with our target setting and to emphasize the dominant road users that drive collision risk.

To ensure the conclusions reflect modern perception systems, we evaluate strong, widely adopted detection backbones: YOLOv8 [18] (a representative high-performance one-stage detector with an established deployment ecosystem), the newer YOLOv11 [19] (a recent upgrade in the YOLO family with improved efficiency/accuracy trade-offs), and RF-DETR [20] (a contemporary DETR-style detector that is competitive with recent transformer-based detectors). Together, they cover both CNN-style real-time detectors and transformer-based detection paradigms, making the study indicative for practical ITS stacks.

Building on our weather editing pipeline, we first synthesize diverse challenging-weather renderings (rain, snow, fog, sandstorm) at multiple CFG-guided appearance grades (including clear) for each curated street scene, forming RoadWeather-34K. To obtain consistent supervision for downstream perception, we start from the original clear-weather images and generate initial bounding-box annotations using a state-of-the-art detector, RF-DETR-l [20], for our target traffic categories. These RF-DETR predictions are treated as pseudo-labels and are manually verified and corrected (e.g., fixing missed objects, removing false positives, and refining box extents) to ensure label reliability on the clear domain. We then propagate the verified labels to the corresponding weather-transferred variants (same underlying scene layout) and conduct an additional round of human inspection to correct potential misalignments or ambiguities introduced by

weather artifacts and partial occlusions. This curation reduces the dataset from 34K to approximately 4K images, yielding a high-quality synthetic strong-weather training subset for downstream robustness enhancement.

We then fine-tune the above pre-trained detectors using only the curated synthetic strong-weather subset and evaluate them on DAWN using DAWN annotations only for scoring (i.e., no DAWN images/labels are used for training). This protocol directly tests whether synthetic weather augmentation can transfer to real challenging-weather conditions and bridge the domain gap in object recognition.

Fig. 11 shows consistent improvements across model families and scales on DAWN. Fine-tuning on our synthetic strong-weather data increases mAP_{50} , $mAP_{50:95}$, and mean matched-box IoU (MBIoU; bbox overlap of matched detections). It also reduces Miss markedly (Miss = $1 - \text{Recall}$ under our matching rule), indicating improved recall and robustness under real challenging weather. For YOLOv8, the gains are $+15.28 \sim +18.65$ mAP_{50} and $+17.68 \sim +22.10$ MBIoU, with a $20.42 \sim 26.01$ -point drop in Miss. YOLOv11 shows comparable gains of $+15.71 \sim +19.51$ mAP_{50} and $+18.40 \sim +21.59$ MBIoU, and Miss decreases by $13.88 \sim 25.69$ points. RF-DETR also improves steadily, with $+4.97 \sim +7.45$ mAP_{50} and $+2.47 \sim +4.64$ MBIoU, while Miss decreases by $3.25 \sim 5.88$ points. Precision remains largely comparable and sometimes slightly decreases due to the recall-oriented shift. The consistent AP gains and Miss reductions show that our synthetic strong-weather appearance spectrum provides effective training signals for real-world challenging-weather

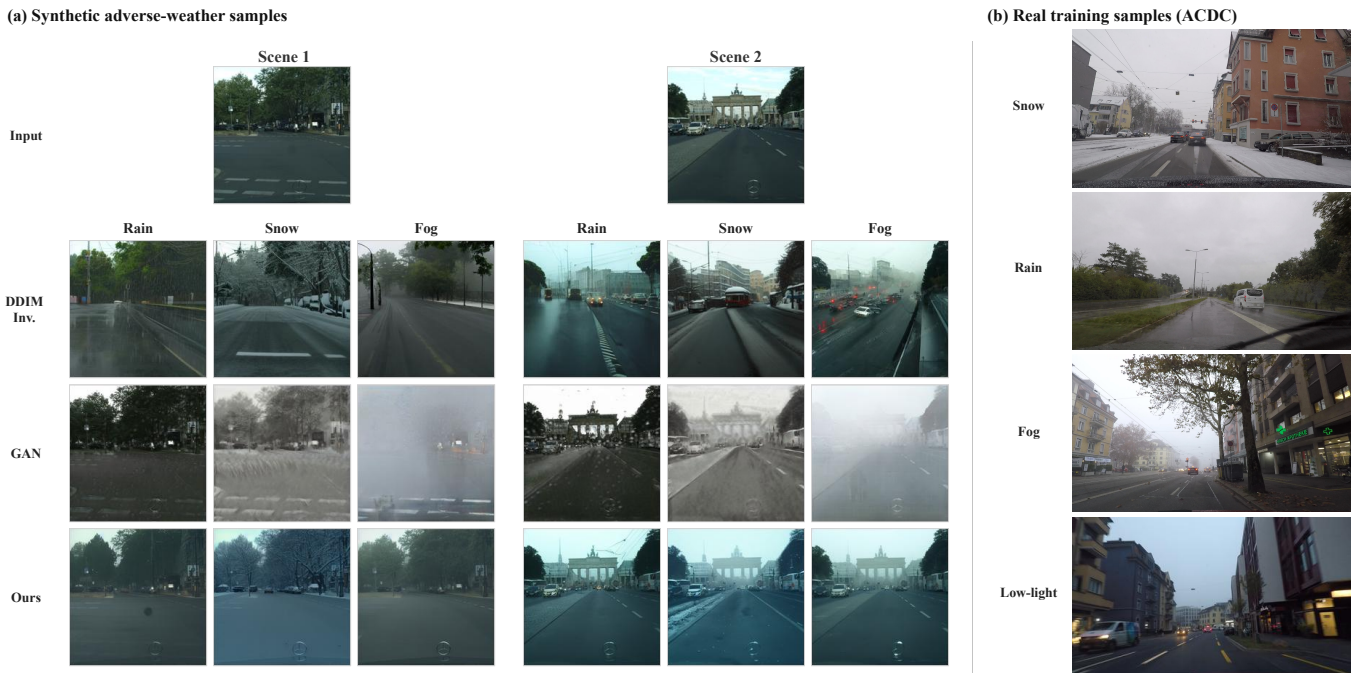


Fig. 12. **Examples of synthetic challenging-weather data and real ACDC samples.** The figure compares Cityscapes clear-weather sources, synthetic rain/snow/fog images generated by different methods, and real ACDC challenging-weather examples. DDIM Inv. denotes the naive DDIM-inversion diffusion baseline, which is included only as a qualitative visualization because its converted images severely damage the original street layout and are not reliable enough for annotation-consistent downstream detector training. GAN and MsGA correspond to the synthetic training sources evaluated in Fig. 11(b) and Table III. The comparison visualizes both the synthetic–real domain gap and the ability of each generator to preserve street layout while approximating real challenging-weather appearance.

object recognition. For the ACDC synthetic-to-real experiment, we construct a stricter protocol from Cityscapes clear-weather sources and real ACDC challenging-weather data. The quantitative comparison contains three training-data settings: *GAN-based weather transfer*, *MsGA-Edit*, and *ACDC-real*. The GAN baseline uses Foggy-CycleGAN for fog synthesis and Weather Effect Generator for rain and snow synthesis. We also generate naive DDIM-inversion outputs for qualitative comparison in Fig. 12. This baseline severely damages the original street geometry and object layout, making the transferred images nearly unusable for annotation-consistent detector training. Therefore, it is excluded from downstream detector training. The first two downstream training settings generate synthetic rain, snow, and fog from the same Cityscapes clear-weather images. This controls the source data and isolates the quality of the generated training data. Each synthetic setting contains 300 rain, 300 snow, and 300 fog images, matching the 900-image scale of the original ACDC challenging-weather training set. GAN-based weather transfer is used as the synthetic downstream baseline. LED-ITS++ is kept as a zero-shot editing comparison because its prompt/mask/schedule-sensitive behavior and weaker batch-level weather consistency make it less suitable for annotation-consistent large-scale detector training under this protocol. The ACDC-real setting directly uses real ACDC challenging-weather training images and therefore serves as a practical upper-bound reference. ACDC test images are used only for final evaluation across rain, snow, and fog.

Specifically, we train YOLO-series variants from scratch under these three settings and evaluate all models on the

TABLE III
ACDC SYNTHETIC-TO-REAL QUANTITATIVE SUMMARY. RESULTS ARE AVERAGED OVER EIGHT YOLO-SERIES SETTINGS IN FIG. 11(B). PARENTHESES REPORT THE MIN–MAX RANGE ACROSS SETTINGS.

Training data	mAP ₅₀ ↑	mAP _{50:95} ↑	MBIoU ↑	Miss ↓
GAN-based	34.73 (27.09–39.06)	20.15 (15.61–23.71)	62.50 (54.81–70.10)	69.09 (66.07–72.94)
MsGA-Edit	47.04 (43.76–50.36)	29.04 (26.62–31.52)	69.70 (66.54–71.28)	60.60 (58.47–63.05)
ACDC-real	60.40 (56.81–64.03)	39.40 (35.67–42.95)	71.66 (69.49–73.24)	46.63 (43.42–49.05)

same real ACDC rain/snow/fog test split. This second downstream task evaluates how closely the generated dataset approaches the real challenging-weather training distribution. Since ACDC-real directly uses real challenging-weather training images, we treat it as the practical upper-bound reference for synthetic data. As shown in Fig. 11(b) and Table III, MsGA-Edit consistently outperforms GAN-based synthetic weather transfer. Averaged over the eight YOLO-series settings, MsGA-Edit improves mAP₅₀ from 34.73 to 47.04 (+12.3), mAP_{50:95} from 20.15 to 29.04 (+8.89), and MBIoU from 62.50 to 69.70 (+7.20). Miss is reduced from 69.09 to 60.60, an 8.48-point drop. Relative to the ACDC-real upper bound, MsGA-Edit reaches 77.9% of its mAP₅₀, 73.7% of its mAP_{50:95}, and 97.3% of its MBIoU on average. The corresponding ratios for GAN-based synthetic data are 57.5%, 51.1%, and 87.2%. These ratios indicate that MsGA-Edit-generated data approximates the real ACDC training distribution to a meaningful extent, although real ACDC data remain stronger in detection accuracy and miss rate. The high MBIoU ratio should be interpreted with this definition in mind: matched-box IoU is computed only over successfully matched detections and is therefore less sensitive to missed detections



Fig. 13. **Generalization to complex urban street scenes.** We evaluate MsGA-Edit on scenes with dense vehicles, buildings, and traffic structures. The method preserves the main road layout and object arrangement under weather transfer, but minor omissions or local errors may still appear in building details and distant small structures.

than AP and Miss. Fig. 12 further provides qualitative examples for the ACDC protocol. The naive DDIM-inversion baseline is included only as a visual reference. It can introduce weather-like appearance, but it severely disrupts the original road geometry, vehicles, and scene boundaries. We therefore did not use its outputs for downstream detector training. Compared with GAN-based translation, MsGA-Edit better maintains these street structures while adding weather-specific appearance. This explains the stronger transfer of detectors trained on MsGA-Edit samples to real ACDC images: the generated data contain stable object/layout supervision and realistic strong-weather degradation.

F. Generalization to Complex and Aerial Urban Scenes

We first test MsGA-Edit on complex urban scenes with dense vehicles, buildings, and traffic infrastructure. As shown in Fig. 13, our method still produces stable weather-transfer results and preserves the dominant street structure under challenging layouts. The road topology, vehicle positions, and building arrangement remain largely consistent with the source image. However, when the scene contains many small structures or far-field details, some building details and distant objects may be omitted or locally distorted. This suggests that the current structural priors are effective for global layout preservation but can still be limited by the resolution and reliability of semantic/edge/depth cues in highly cluttered areas.

We also evaluate cross-view generalization on UAVid [52], which provides high-resolution UAV imagery of urban street scenes. Fig. 14 shows that MsGA-Edit can be transferred to aerial urban scenes without training. The structural preservation remains effective: roads, buildings, vehicles, and vegetation regions are largely kept in their original positions.

Nevertheless, several limitations become visible in this viewpoint. Fog sometimes appears spatially uneven, and when translating scenes into winter or snowy conditions, vegetation regions can remain unrealistically green and the snow-specific appearance may be insufficient. These results indicate that MsGA-Edit generalizes beyond ground-view driving scenes in terms of structure protection, but aerial-view weather realism still requires stronger priors for vegetation, seasonal change, and large-area atmospheric consistency.

VI. CONCLUSION

We presented MsGA-Edit, a training-free diffusion framework for layout-preserving weather-effect transfer in driving scenes. In this formulation, classifier-free guidance ω controls apparent weather strength. Structure-Aware Guidance, masked MsGA, and DVI preserve road topology, object positions, and local street-scene structure under stronger weather prompts. Experiments show that this layout-preserving design improves the trade-off between weather appearance transfer and image/structure preservation over representative zero-shot editors. The generated challenging-weather data also benefit downstream detection on DAWN and ACDC.

RoadWeather-34K further demonstrates the practical value of this formulation as a synthetic training resource. In our ACDC protocol, detectors trained or fine-tuned with MsGA-Edit data transfer better to real challenging-weather benchmarks than detectors trained with GAN-based synthetic augmentation. MsGA-Edit reaches 77.9% of the ACDC-real mAP₅₀, 73.7% of the ACDC-real mAP_{50:95}, and 97.3% of the ACDC-real MBIoU on average. Because MBIoU is computed only over successfully matched boxes, this high ratio mainly reflects box-alignment quality among detected objects and should be read together with AP and Miss. These results



Fig. 14. **Generalization to UAVID aerial urban scenes.** We apply MsGA-Edit to UAVID, a high-resolution UAV dataset for urban street scenes. The results show strong structure preservation from an aerial viewpoint, while fog may be spatially uneven and winter/snow transfer can leave vegetation regions overly green, indicating insufficient modeling of aerial-view snow coverage and seasonal vegetation changes.

support the use of layout-consistent synthetic weather data for robustness enhancement. They also show that fully real challenging-weather data remain the upper-bound reference.

Limitations and Future Work. MsGA-Edit is not a calibrated meteorological simulator: CFG-based appearance grades are not mapped to measurable quantities such as rainfall rate, snowfall rate, particle-size distribution, aerosol density, or fog visibility. DVI introduces relative depth-aware appearance weighting, but it is not a physical scattering model. The method also operates on single images, depends on the quality of semantic/edge/depth cues, and incurs additional cost from DDIM inversion and feature caching. Future work will calibrate generated weather appearance against measurable physical parameters or sensor metadata, extend the method to temporally consistent video, and distill the two-stage pipeline into faster few-step solvers for practical deployment.

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